

Hybrid 2D–3D Reconstruction of the 2011 Tohoku Tsunami for Coastal-Defence Assessment

Confidence in numerical assessments of coastal-defence performance under tsunami loading is strengthened by validation against independent observations of tsunami propagation and coastal impact.

Here, a one-way coupled 2D–3D framework is developed to reconstruct the 2011 Tohoku tsunami from coseismic generation to coastal impact.

Coseismic seafloor displacement defines the initial free-surface disturbance for a 2D nonlinear shallow-water model simulating propagation through a source-to-coast corridor.

Bathymetric and topographic datasets define the terrain, while absorbing layers and radiation conditions reduce spurious reflections at truncated open-ocean boundaries.

Extracted free-surface-elevation and depth-averaged velocity histories are mapped to time-dependent inlet conditions for a 3D unsteady Reynolds-averaged Navier–Stokes model with volume-of-fluid interface capture, simulating inundation and interaction with rigid defences and buildings.

Model performance is evaluated against observed arrival times, offshore wave-forms, run-up heights and inundation extents using errors in timing, amplitude and spatial extent.

Expected outcomes are close quantitative agreement with event observations, physically consistent regional-to-local forcing, and resolved flow fields and hydrodynamic loads.

The resulting event-validated framework will enable comparative assessment of defence configurations under the historical event and systematically varied source scenarios.

Subsequent development will incorporate fluid–structure interaction and damage modelling to assess defence deformation, damage progression and failure margins under extreme tsunami loading.